ABDURAHIMAN THOMMANGADAN

AI Full Stack Engineer | Python, Node.js, React, Generative AI & AI Agents

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PROFESSIONAL SUMMARY

AI Full Stack Engineer and solution architect with 7+ years in Node.js and React and 4 years of Python (FastAPI, Django), building scalable, secure products from scratch on MERN, NestJS, MongoDB and PostgreSQL, including Samsung.com global eCommerce (Samsung SDS) and enterprise SaaS (Infosys). Technical Lead for full stack and Generative AI at Voizzit since 2024, delivering CADBEATER, an AI estimator system for fenestration and facade projects, end to end: Python LangGraph agents, OpenAI, Anthropic and OpenRouter LLMs, vLLM inference, LLMOps, FastAPI and React. Strong in system design, Clean (Onion) Architecture, distributed and event-driven systems, scaling, RBAC security, AWS, CI/CD and AI-driven SDLC team leadership.

TECHNICAL SKILLS

Generative AI & Agents: LangGraph, LangChain, Multi-Agent Systems, AI Agent Development, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Context & Memory Management, Prompt Engineering, Large Language Models (LLMs), LLM Inference Engineering

LLMs, Inference & LLMOps: OpenAI, Anthropic (Claude), OpenRouter, Hugging Face, Structured Outputs (Constrained Decoding), vLLM, Quantization, Model Routing, Response Caching, Langfuse, Evaluation Pipelines, Cost & Latency Monitoring

Backend: Python (4 years), FastAPI, Django, Node.js, Express.js, NestJS, REST APIs, gRPC, WebSocket, Server-Sent Events (SSE), ORM / ODM (Sequelize, TypeORM, Mongoose)

Architecture & System Design: Solution Architecture, System Design, Clean (Onion) Architecture, SOLID & Design Patterns, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Microservices, Event-Driven Architecture (Kafka, RabbitMQ), Distributed Systems (MongoDB, PostgreSQL), Horizontal & Vertical Scaling, Caching (Redis), API Gateway, Asynchronous Workers & Queues, High Availability

Security: Role-Based Access Control (RBAC), JWT & OAuth 2.0 Authentication, OWASP Top 10, Secure API Design, Rate Limiting, Encryption in Transit & at Rest, Audit Logging

Databases & Messaging: MongoDB, PostgreSQL, MySQL, Redis, Qdrant, Apache Kafka, RabbitMQ

Frontend & Mobile: React.js, Redux Toolkit, TanStack Query, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Three.js, Angular, Flutter (Android, iOS, Web), React Native

Cloud & DevOps: AWS (EC2, S3, SQS), Docker, Kubernetes, GitHub Actions, CI/CD, ELK Stack, RunPod

AI-Driven SDLC & Testing: Agentic Coding (Claude Code), Custom Module-Level & Project-Level Agent Skills, AI-Assisted Code Review, AI Development Lifecycle, Jest, Supertest, Playwright

Languages: Python (4 years), JavaScript, TypeScript, Dart, SQL

PROFESSIONAL EXPERIENCE

Voizzit Information Technology LLC, Dubai, UAE — Sep 2024 – Present

Technical Lead — Full Stack & Generative AI

Flagship project: CADBEATER

- Led CADBEATER from requirements through production across the Python multi-agent AI backend, FastAPI services, MongoDB data layer and React frontend. Its estimator generates cost estimates, bills of quantities (BOQ), bills of materials (BOM) and CAD drawings from requirements, joinery details or floor plans, reducing manual estimating effort from four hours to ten minutes (about 96% less time).
- Built the multi-agent AI layer in Python: LangGraph agents (ReAct, Reflexion, Plan-and-Execute), RAG over Qdrant, Model Context Protocol (MCP) tool integration, and short- and long-term agent memory with context compression.
- Integrated LLMs from OpenAI, Anthropic (Claude) and OpenRouter alongside a self-hosted vision-language model served with vLLM, with task-based model routing, caching and structured outputs through constrained decoding.
- Engineered for scale and reliability: event-driven Python (FastAPI) services on Redis Streams, GPU-isolated inference workers, and real-time SSE streaming to the React CAD frontend (Redux Toolkit, TanStack Query, Three.js).
- Secured the platform with role-based access control (RBAC), role- and team-scoped data access, audit logging and human-in-the-loop approval gates.

- Owned quality and LLMOps: Langfuse tracing, prompt versioning, evaluation pipelines, cost and latency monitoring, code reviews and an AI-driven SDLC (agentic coding with Claude Code, custom module- and project-level agent skills), and authored the architecture reference adopted across engineering and product.

  Tech stack: Python, FastAPI, LangGraph, MCP, Qdrant, vLLM, MongoDB, Redis Streams, React.js

Senior Backend Engineer — SaaS Platforms

- Architected VData, a distributed voter-data platform, on NestJS with Clean (Onion) Architecture, separating domain, application and infrastructure layers, with TypeORM over PostgreSQL.
- Scaled VData horizontally and vertically on AWS to enable fast, near-instant search across 30 million voter records; supported the platform with ELK analytics, CI/CD and automated API test suites.
- Built TechBuddy LMS (admin CMS, student web, React Native TV apps) on NestJS and PostgreSQL, with AI-generated quizzes, offline content export, secure authentication and rate limiting.

  Tech stack: NestJS, TypeORM, PostgreSQL, Python, AWS, ELK Stack, React Native

Project: AutoFleet — Car Rental Booking & ERP Platform

- Built a car rental booking and ERP platform covering the end-to-end booking flow, fleet management, reservations, CRM and HRM modules.
- Delivered a cross-platform Flutter mobile app for Android, iOS and web, and handled release management and ongoing maintenance on the Google Play Store and Apple App Store.
- Built the Python (FastAPI) backend and React.js web platform on MongoDB and AWS, with Playwright-based automation for third-party integrations.

  Tech stack: Python (FastAPI), React.js, Flutter (Dart), MongoDB, AWS, Playwright

Samsung SDS India, Chennai, India — Feb 2024 – Sep 2024

Backend Engineer — Samsung.com Global eCommerce

- Built and operated independently deployable Node.js microservices for Samsung.com global eCommerce, supporting autoscaling during high-traffic sales events; used Kafka and RabbitMQ for event processing and PostgreSQL for transactional data.
- Built secure logistics and user-interaction APIs with Elasticsearch logging and monitoring for end-to-end traceability.

Infosys Limited, Bangalore, India — Jun 2022 – Feb 2024

Associate Consultant — Enterprise SaaS

- Delivered full-stack features and high-availability microservices for Solargraf, a solar-design SaaS with 3D design and financial modules (Node.js, Express.js, React.js, Sequelize, PostgreSQL).
- Built and tested microservices behind an API Gateway with gRPC, and took part in architecture reviews with cross-functional product teams.

Kimball Electronics India Pvt. Ltd., Trivandrum, India — Nov 2020 – Jun 2022

Software Engineer — Industrial IoT

- Built the Alerting, Visualization and Data Acquisition modules of EMTAB, an Industry 4.0 IoT predictive-maintenance platform (Node.js, React.js, Express.js, Sequelize, MongoDB, MySQL).
- Integrated machine-learning anomaly detection, written in Python, into real-time shop-floor sensor pipelines, and wrote Playwright test automation.

  Tech stack: Node.js, React.js, MongoDB, MySQL, Python (ML anomaly detection), Playwright

Nucore Software Solutions, Calicut, India — Apr 2019 – Nov 2020

Backend Developer — Travel Accounting SaaS

- Designed and built the ERP accounting engine, microservice APIs and business logic for TRAACS, a back-office SaaS for travel agencies (Node.js, Express.js, Angular, PostgreSQL, MongoDB).

EDUCATION

B.Sc. Computer Science, Farook College, Kozhikode (University of Calicut) — 2016 – 2019

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI (Andrew Ng)
- Deep Learning — Codebasics